

David Kennedy

Final: Manufacturing Pharmaceuticals using concepts from Transport Phenomena II

(Includes original problems: do not repost anywhere without permission)

Marked out of 100; contribution to the final grade varies

Edited Appendix B from the Transport Phenomena text (by BSL) is available for reference. Handwritten formula sheet will be graded too and is worth five points (at maximum). Assume that the time in minutes needed for solving a problem is comparable to (or less than) the maximum score possible on that problem.

Problem 1: For a given concentration profile (determined by an chemical engineer intern at a

pharmaceutical company), $\frac{C}{C_0} = \sin\left(-\pi x^2 / 4D_{AB}t\right) - \left(\frac{x^2}{4D_{AB}t}\right) \left(1 - \operatorname{erf} \frac{x}{\sqrt{4D_{AB}t}}\right)$, where C_0 and D_{AB}

are both constant and known (6+4+5=15), find:

(a) the local mass flux in x-direction at $x = 0$.

(b) the concentration at $x = 0$ and at $x^2 = 4D_{AB}t$.

(c) $\frac{Dc}{Dt}$.

$$a) N = -D_{AB} \frac{\partial C}{\partial x}$$

$$= -D_{AB} \frac{\partial \frac{C}{C_0}}{\partial x} * C_0$$

$$\left. N \right|_{x=0} = 0$$

$$b) C @ x=0$$

$$\frac{C}{C_0} = 0 - 0$$

$$C = 0 \text{ at } x=0$$

$$C @ x^2 = 4D_{AB}t$$

$$\frac{C}{C_0} = \sin(-\pi) - 1 * (1 - \operatorname{erf} 1)$$

$\operatorname{erf} 1 \approx 0.8$: I recall

$$C = 0.2C_0 \text{ at } x^2 = 4D_{AB}t$$

$$\text{need } \frac{\partial \frac{C}{C_0}}{\partial x}$$

$$\frac{\partial \frac{C}{C_0}}{\partial x} = \frac{-2x\pi}{4D_{AB}t} \cos\left(\frac{-\pi x^2}{4D_{AB}t}\right) +$$

$$\left(\frac{-2x}{4D_{AB}t}\right) \left(1 - \operatorname{erf} \frac{x}{\sqrt{4D_{AB}t}}\right)$$

$$- \left(\frac{x^2}{4D_{AB}t}\right) \left(-\frac{2}{\sqrt{\pi}} \exp\left(-\frac{x^2}{4D_{AB}t}\right) * \frac{1}{\sqrt{4D_{AB}t}}\right)$$

at $x=0$ all terms go to zero

$$c) \frac{Dc}{Dt} = \frac{dc}{dt} + v_x \frac{dc}{dx} + v_y \frac{dc}{dy} + v_z \frac{dc}{dz}$$

$$= C_0 \frac{d\frac{C}{C_0}}{dt} + v_x C_0 \frac{d\frac{C}{C_0}}{dx}$$

$$\text{need } \frac{d\frac{C}{C_0}}{dt}$$

$$\frac{d \frac{c}{c_0}}{d \psi} = \frac{-\pi x^2}{4 D_{AB} t} h(t) \cos \frac{-\pi x^2}{4 D_{AB} t}$$

$$- \frac{x^2}{4 D_{AB}} h(t) \left(1 - \operatorname{erf} \frac{x}{\sqrt{4 D_{AB} t}} \right)$$

$$- \frac{x^2}{4 D_{AB} t} \left(-\frac{2}{\sqrt{\pi}} \exp \left(-\frac{x^2}{4 D_{AB} t} \right) \frac{x}{\sqrt{4 D_{AB}}} * \left(-\frac{1}{2} \right) t^{-3/2} \right)$$

$$\begin{aligned} \frac{D_c}{D_t} = C_0 & \left[\frac{-2 \times \pi}{4 D_{AB} t} \cos \left(\frac{-\pi x^2}{4 D_{AB} t} \right) - \left(\frac{2x}{4 D_{AB} t} \right) \left(1 - \operatorname{erf} \left(\frac{x}{\sqrt{4 D_{AB} t}} \right) \right) - \frac{x^2}{4 D_{AB} t} \left(-\frac{2}{\sqrt{\pi}} \exp \left(\frac{-x^2}{4 D_{AB} t} \right) * \frac{1}{\sqrt{4 D_{AB} t}} \right) \right. \\ & \left. + 2x \left[\frac{-\pi x^2}{4 D_{AB}} h(t) \cos \frac{-\pi x^2}{4 D_{AB} t} - \frac{x^2}{4 D_{AB}} h(t) \left(1 - \operatorname{erf} \frac{x}{\sqrt{4 D_{AB} t}} \right) - \frac{x^2}{4 D_{AB} t} \left(-\frac{2}{\sqrt{\pi}} \exp \left(-\frac{x^2}{4 D_{AB} t} \right) \frac{x}{\sqrt{4 D_{AB}}} * \left(-\frac{1}{2} \right) t^{-3/2} \right) \right] \right] \end{aligned}$$

Problem 2: Heat Transfer and Fever: Designer Cold Patches

The forehead of a person running a high fever is at temperature T_0 . A cold wet towel of thickness d and initial temperature $T_1 < T_0$ is placed on the forehead at time $t = 0$. Assuming the body-temperature is maintained constant, find the time-dependent temperature profile within the towel. Also determine the heat flux into the towel. For a wet towel of area A , how much heat is transferred per second from the body to the towel? (You can probably help a pharmaceutical company design more efficient cold patches using a better understanding of heat transfer concepts involved, and the analysis you write down here can be used for picking the right material or right thickness for wet towels). (12+4+4)



- time dependent temperature profile within towel,
- heat flux into towel
- heat transfer per second for area A

Energy Balance

$$In - Out + \underset{\downarrow}{G_{gen}} = Accu$$

$$In = qA \text{ at interface } x=0$$

$$Out = qA \text{ at } x=d$$

not sure about Accumulation

Assume d is effectively infinite

use error function solution

of the form

$$\frac{T - T_1}{T_0 - T_1} = C_1 + C_2 \operatorname{erf} \frac{x}{\sqrt{4 \frac{k}{\rho C_p} t}}$$

$$\boxed{\frac{T - T_1}{T_0 - T_1} = 1 - \operatorname{erf} \frac{x}{\sqrt{4 \frac{k}{\rho C_p} t}}}$$

Apply B.C.2

$$1 = C_1 + 0$$

Apply I.C

$$0 = 1 + C_2 \operatorname{erf}(\infty)$$

$$C_2 = -1$$

Heat flux $q = -k \frac{\partial T}{\partial x}$

define $\phi = \frac{T - T_1}{T_0 - T_1}$

$$\frac{\partial \phi}{\partial x} = \frac{\partial T}{\partial x} \frac{\partial \phi}{\partial T}, \quad \frac{\partial \phi}{\partial T} = \frac{1 - T_1}{T_0 - T_1}$$

$$q = -k \left(\frac{1 - T_1}{T_0 - T_1} \right) \frac{\partial \phi}{\partial x}$$

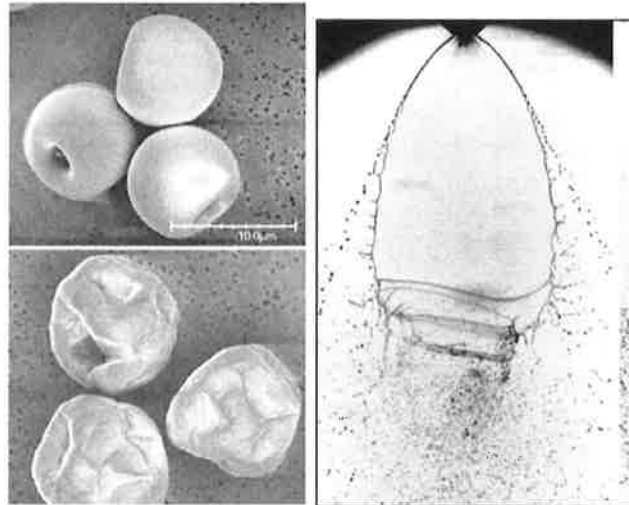
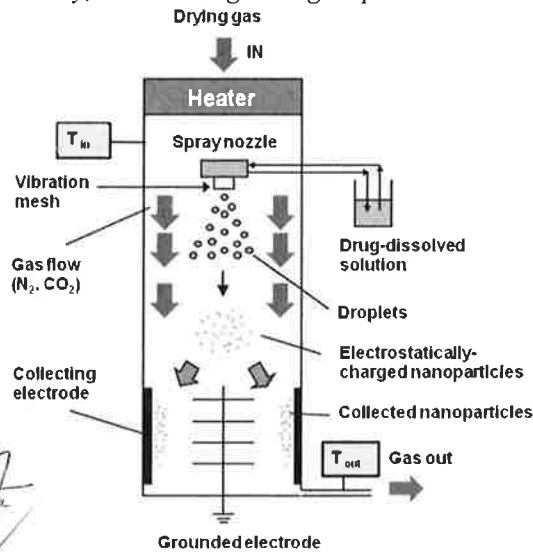
$$\frac{\partial \phi}{\partial x} = \frac{-2}{\sqrt{\pi}} \exp\left(-\frac{x^2}{4 \frac{k}{\rho C_p} t}\right) \cdot \frac{1}{\sqrt{4 \frac{k}{\rho C_p} t}}$$

$$q = -k \left(\frac{1 - T_1}{T_0 - T_1} \right) \left(-\frac{2}{\sqrt{\pi}} \exp\left(-\frac{x^2}{4 \frac{k}{\rho C_p} t}\right) \frac{1}{\sqrt{4 \frac{k}{\rho C_p} t}} \right)$$

Heat transfer for area A

$$\text{Heat} = q * A = -A k \left(\frac{1 - T_1}{T_0 - T_1} \right) \left(-\frac{2}{\sqrt{\pi}} \exp\left(-\frac{x^2}{4 \frac{k}{\rho C_p} t}\right) \frac{1}{\sqrt{4 \frac{k}{\rho C_p} t}} \right)$$

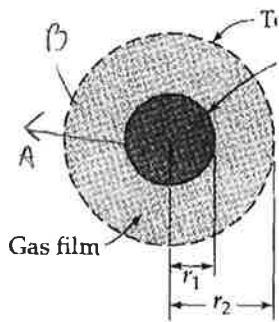
Problem 3. Spray-dried Pharmaceuticals: *Spray-drying process (see the schematic below) is commonly used in the pharmaceutical and food industries for making particles or powders (including dry milk, flavored coffee, etc.). In a typical process, the drug-dissolved solution is through a spray nozzle. The fluid atomizes as it issues out of the nozzle, and the resulting droplets dry out in presence of a drying gas. Finally, the resulting 'charged' particles are collected as shown. (25+25+10=60)*



(Figures credits: *Pharm. Res.* 25(5), 2008 and Villiermaux, *Ann. Rev. Fluid Mech.* 2007)

(a) Assume that a hypothetical spherical stagnant film of gas B surrounds each falling droplet made of fluid A. Assume that a steady-state diffusion of A occurs through the stagnant film of gas B, and the solubility of gas B in liquid A is negligible. Assume that the gas-phase concentration of A, expressed as mole fraction is x_{A1} at the liquid-gas interface (r_1) and the concentration just outside the stagnant film is maintained at x_{A2} . Assume gases A and B are ideal and assume that change in drop size can be neglected. Derive the expressions for diffusion through a spherical shell to obtain both the concentration profile and molar flux. (25)

Mass Balance $I_n - O_{out} + G_{gen} = A_{gen}$
no rxn steady state



$$\frac{d(r^2 N_A)}{dr} = \frac{dr^2}{dr} N_A + \frac{dN_A}{dr} r^2$$

$$N_A = -c D_{AB} \frac{dx_A}{dr} + x_A (N_A + N_B)$$

Stagnant

$$N_A = -\frac{c D_{AB}}{1-x_A} \frac{dx_A}{dr} = -\frac{c D_{AB}}{1-x_A} \frac{(x_{A2} - x_{A1})}{(r_2 - r_1)}$$

$$I_n = \cancel{4\pi r_1^2} S_1 N_A|_{r_1}$$

need surface area of sphere
call it S

$$S = S_1 \text{ at } r_1 \text{ and } S = S_2 \text{ at } r_2$$

$$O_{out} = S_2 N_A|_{r_2}$$

$$S_1 N_A|_{r_1} - S_2 N_A|_{r_2} = 0$$

S is a function of r^2 of the form $r^2 + a$
divide by 'a'

$$(r^2 N_A)|_{r_1} - (r^2 N_A)|_{r_2} = 0 \rightarrow \frac{dr^2 N_A}{dr} = 0$$

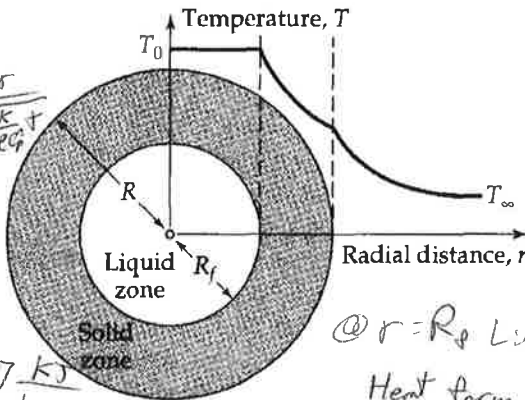
divide by $r_2 - r_1$ and lim $(r_2 - r_1) \rightarrow 0$

(b) **Freezing Drops.** Rather than drying, some pharmaceutical are produced by freezing the droplets. In such a scenario, the droplets leave the nozzle only slightly above their melting point, and drying gas plays the role of a colder outside medium. Estimate the time t_f required for a drop of radius R to freeze completely, if the drop is initially at its melting point T_0 and the surrounding air is at T_1 . Heat is lost from the drop according to the Newton's law of cooling, with constant heat-transfer coefficient h . Assume that there is no volume change in the solidification process. Solve the problem by using a **quasi-steady-state method**.

(HINT: (a) First solve the steady-state heat conduction problem in the solid phase in the region between $r = R_f$ (the liquid-solid interface) and $r = R$ (the solid-air interface). Let k be the thermal conductivity of the solid phase. Then find the radial heat flow Q across the spherical surface at $r = R$. (b) Assuming H be the latent heat of freezing per unit mass, and using unsteady-state energy balance, find the time that drop takes to solidify). (25=15+10)

Estimate with error function
Solution

$$\frac{T - T_\infty}{T_0 - T_\infty} = 1 - \operatorname{erf} \frac{r}{\sqrt{4 \frac{k}{\rho C_p} t}}$$



$$q = \frac{kT}{\text{Area} \cdot s} = \frac{kT}{\text{Area} \cdot s} \cdot \frac{\text{Vol}}{\text{Vol}} \cdot \frac{\text{dist}}{s}$$

$$\frac{kT}{\text{Area} \cdot s} = \frac{kT}{\text{Area} \cdot s} \cdot \frac{\text{Vol}}{\text{Vol}} \cdot \frac{\text{dist}}{s}$$

$$\frac{k}{\rho H} \frac{\text{KT}}{\text{m}^3 \cdot \text{K}} = \frac{\text{m}^2}{\text{s}} \text{ diffusivity}$$

$$r = -C_1^2 t + C_2$$

$$T = T_\infty + (r - R) \frac{C_1^2}{\alpha}$$

Energy Balance

$$\text{In} - \text{Out} + G_{\text{gen}} = A g_{\text{con}} \quad \text{steady state}$$

$$\text{I.C. } T = T_0 \text{ at } t = 0$$

$$\text{B.C.1: } T = T_\infty \text{ at } r \rightarrow \infty$$

$$\text{B.C.2: } T = T_\infty \text{ at } r = R \text{ for } t > 0$$

@ $r = R_f$ Liquid-Solid interface

Heat from liquid phase goes to the solid phase

$$q = \frac{k dT}{dr} = \rho H \frac{dr}{dt}$$

@ $r = R$ Solid-gas interface

$$q = -k \frac{dT}{dr}$$

$$\text{I.C. } T = T_0 \text{ at } r = R, t = 0$$

P.O. Guess

$$\frac{dr}{dt} = -\alpha \frac{dT}{dr} = -C_1^2$$

$$\frac{dr}{dt} = -\alpha \frac{dT}{dr}$$

$$\frac{dr}{dt} = -\alpha \frac{dT}{dr}$$

$$\frac{dr}{dt} = -C_1^2$$

$$r = -C_1^2 t + C_2$$

dimensionalize

$$\text{Define } \phi = \frac{r}{R}$$

$$\bar{T} = \frac{T - T_\infty}{T_0 - T_\infty}$$

$$\bar{T} = \frac{r}{\sqrt{4 \frac{k}{\rho C_p} t}}$$

$$R \frac{d\phi}{dt} = -\alpha \frac{d\bar{T}}{d\phi} = -C_1^2$$

apply B.C.2

$$T = \frac{C_1^2 r}{\alpha} + C_3$$

$$T_\infty = \frac{C_1^2 R}{\alpha} + C_3$$

$$C_3 = T_\infty - \frac{C_1^2 R}{\alpha}$$

$$T = \frac{C_1^2 r}{\alpha} + T_\infty - \frac{C_1^2 R}{\alpha} = T_\infty + (r - R) \frac{C_1^2}{\alpha}$$

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(c) Mass transfer from a droplet: (All quantities in this problem are in SI units. Appropriate units must accompany your final numerical answer). The droplets of diameter of 0.01 (in SI units) fall through the gas B with velocity of 4 SI units. The drying gas has viscosity 4×10^{-4} units, thermal conductivity of 0.025, density of 1.1, specific heat per unit mass of 1000 and the diffusion coefficient is $1/10^9$ SI units. **Give a simple estimate for mass transfer coefficient relevant for this problem, useful for estimating mass flux out of drying droplets. (10)**

On Heroic Effort

A hero ventures forth from the world of common day into a region of supernatural wonder: fabulous forces are there encountered and a decisive victory is won: the hero comes back from this mysterious adventure with the power to bestow boons on his fellow man.

Joseph Campbell: A Hero with a Thousand Faces.

Combined Flux Expressions

Momentum:

$$\underline{\underline{\phi}} = \rho \underline{\underline{v}} \underline{\underline{v}} + \rho \underline{\underline{\delta}} - \underbrace{\mu (\nabla \underline{\underline{v}} + (\nabla \underline{\underline{v}})^T)}_{\underline{\underline{\tau}}}$$

Energy:

$$\underline{\underline{e}} = \rho (\hat{U} + \frac{1}{2} \underline{\underline{v}}^2) \underline{\underline{v}} - K \nabla T + [\underline{\underline{\tau}} \cdot \underline{\underline{v}}] \quad \text{molecular work Flux}$$

$$= \rho (\hat{H} + \frac{1}{2} \underline{\underline{v}}^2) \underline{\underline{v}} - K \nabla T + [\underline{\underline{\tau}} \cdot \underline{\underline{v}}]$$

Mass:

$$n_A = \underbrace{\rho u_A \underline{\underline{v}}}_{\text{Convective terms}} - \underbrace{\rho D_{AB} \nabla u_A}_{j_A}$$

Error Function $\text{erf}(0) = 0$
 $\text{erf}(2) = 0.99$

$$\text{erf } x = \frac{\int_0^x \exp(-\bar{x}^2) d\bar{x}}{\int_0^\infty \exp(-\bar{x}^2) d\bar{x}} = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-\bar{x}^2) d\bar{x}$$

Gamma Function $\frac{d}{dx} \text{erf } x = \frac{2}{\sqrt{\pi}} \exp(-x^2) \frac{dx}{dx}$

$$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx$$

$$\Gamma(n) = \int_0^\infty (\ln \frac{1}{x})^{n-1} dx$$

$$\Gamma(n+1) = \int_0^\infty \exp(-x^{1/n}) dx$$

If using molar average velocity $\underline{\underline{v}}^*$ instead mass average velocity $\underline{\underline{v}}$ the z component of mol flux is

$$N_{Az} = x_A (N_{Az} + N_{Bz}) - c D_{AB} \frac{\partial x_A}{\partial z}$$

Dimensionless Groups

(l_0 and v_0 are characteristic length and velocity respectively)

$Re = l_0 v_0 \rho / \mu$	$Pr = \hat{c}_p \mu / K$	$Sc = \mu / (\rho D_{AB})$
$Ra = Gr Pr$	$Gr = g \beta l_0^3 \Delta T / v^2$	$Gr_w = g \beta l_0^3 \Delta u_A / v^2$
$Nu = h l_0 / K$	$Pe = Re Pr$	$Pe_{AB} = Re Sc$
$Sh = K l_0 / D_{AB}$	$j_H = Nu / Re Pr^{1/3}$	$j_D = Sh / Re Sc^{1/3}$

Analogies Among Heat and Mass Transfer at low Mass Transfer Rates

	Profile	Diffusivity	Effect of Density	Flux	Transfer Rate	Transfer Coefficient	Chilton-Colburn j-factors
Heat transfer	T	$\alpha = \frac{K}{\rho \hat{c}_p}$	$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_p$	$\underline{\underline{q}}$	Q	$h = \frac{Q}{A \Delta T}$	$j_H = Nu Re^{-1} Pr^{-1/3}$ $= \frac{h}{\rho \hat{c}_p v_0} \left(\frac{c_p \mu}{K} \right)^{2/3}$
Binary mass transfer	x_A	D_{AB}	$\bar{\beta} = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial x_A} \right)_p$	$\underline{\underline{J}}_A^*$	$\frac{W_{A0} - x_{A0}(W_{A0} + W_{B0})}{x_{A0}(W_{A0} + W_{B0})}$	$K_x = \frac{\text{Transfer rate}}{A \Delta x_A}$	$j_D = Sh Re^{-1} Sc^{-1/3}$ $= \frac{K_x}{c v_0} \left(\frac{\mu}{\rho D_{AB}} \right)^{2/3}$
Molar limits							

$$D_{AB} \text{ (cm}^2/\text{s)} \quad G/G: 10^{-1}; L/L: 10^{-5}; G/S: 10^{-10}; S/S: 10^{-20} \text{ to } 10^{-30}$$

For D: Diffusion Through Stagnant Film

$$N_{A2} = - \frac{c D_{AB}}{1-x_A} \frac{dx_A}{dz} \quad \left(\frac{1-x_A}{1-x_{A1}} \right) = \left(\frac{1-x_{A2}}{1-x_{A1}} \right)^{\left(\frac{z-z_1}{z_2-z_1} \right)}$$

$$x_{B, \text{avg}} = \frac{x_{B2} - x_{B1}}{\ln(x_{B2}/x_{B1})}$$

logarithmic mean

For evaporation

$$N_{A2}|_{z=z_1} = \frac{c D_{AB}}{(z_2-z_1)(x_{B1})_{\ln}} (x_{A1} - x_{A2})$$

If there is a rxn, Flux depends on stoichiometry, For Example, $2A \rightarrow B$ in packed bed of cat. pellets

$$N_{B2} = -\frac{1}{2} N_{A2}$$

$$N_{A2} = - \frac{c D_{AB}}{1-\frac{1}{2}x_A} \frac{dx_A}{dz}$$

$$\left(1 - \frac{1}{2}x_A \right) = \left(1 - \frac{1}{2}x_{A0} \right)^{1-(z/S)}$$

S is thickness of stagnant

film around a cat. pellet

$$N_{A2} = \frac{2c D_{AB}}{S} \ln \left(\frac{1}{1-\frac{1}{2}x_{A0}} \right)$$

Generalized Procedure

Conservation Principle + Flux Equation $\xrightarrow{\text{Insert Flux Law}}$

Second Order D. Differential Equation

Solve $\rightarrow$

May need

to de-dimensionalise,

use Appendix C,

Separation of variables

or combination of variables.

use Initial and Boundary

Conditions.

using results, solve

for desired information

(averages,

max/min, etc.)

Mass Transfer for short Duration

$$\frac{C_A}{C_{A0}} = \text{erfc} \frac{y}{\sqrt{4D_{AB}t}}$$

penetration depth $\delta = 4\sqrt{D_{AB}t}$

Da^{II} Second Dankohler Number

$$Da^{\text{II}} = \frac{k_1'' S}{D_{AB}} \quad \frac{\text{rxn rate}}{D: \text{ diffusivity}}$$

heat transfer $Q(t) = \int_0^L \int_0^{2\pi} \left(+k \frac{\partial T}{\partial r} \Big|_{r=R} \right) R d\theta dz$

mass transfer $W_{A0}(t) - x_{A0}(W_{A2}(t) + W_{B0}(t)) = \int_0^L \int_0^{2\pi} \left(+c D_{AB} \frac{\partial x_A}{\partial r} \Big|_{r=R} \right) R d\theta dz$

heat transfer $h_1(t) = \frac{1}{\pi D L (T_0 - T_1)} \int_0^L \int_0^{2\pi} \left(+k \frac{\partial T}{\partial r} \Big|_{r=R} \right) R d\theta dz$

mass transfer $k_{x1}(t) = \frac{1}{\pi D L (x_{A0} - x_{A1})} \int_0^L \int_0^{2\pi} \left(+c D_{AB} \frac{\partial x_A}{\partial r} \Big|_{r=R} \right) R d\theta dz$

heat transfer $Nu_1(t) = \frac{h_1 D}{k} = \frac{1}{2\pi L D} \int_0^L \int_0^{2\pi} \left(- \frac{\partial T}{\partial r} \Big|_{r=\frac{1}{2}} \right) d\theta dz$

mass transfer $Sh_1(t) = \frac{k_{x1} D}{c D_{AB}} = \frac{1}{2\pi L D} \int_0^L \int_0^{2\pi} \left(- \frac{\partial x_A}{\partial r} \Big|_{r=\frac{1}{2}} \right) d\theta dz$

$$\tilde{r} = \frac{r}{D} \quad \tilde{z} = \frac{z}{D} \quad \tilde{T} = \frac{(T - T_0)}{(T_1 - T_0)} \quad \tilde{x}_A = \frac{(x_A - x_{A0})}{(x_{A1} - x_{A0})}$$

Boussinesq approx

$$\rho(T, w_A) \approx \bar{\rho} - \bar{\rho} \bar{\beta} (T - \bar{T}) - \bar{\rho} \bar{\gamma} (w_A - \bar{w}_A)$$

$$\bar{\gamma} = -(1/\bar{\rho}) \left(\frac{\partial \rho}{\partial w_A} \right) \quad \bar{\beta} = -(1/\bar{\rho}) \left(\frac{\partial \rho}{\partial T} \right)_p$$

Typical Orders of Magnitude for Heat Transfer Coefficients

System	h ($\text{W/m}^2 \cdot \text{K}$)
Free convection	3-250
Gases	1-25
Liquids	100-600
Boiling water	1000-20,000
Forced Convection	
Gases	10-100
Liquids	50-500
Water	500-10,000
Condensing vapors	1000-100,000